new proof of the Frobenius reciprocity and to some analogies between induction and restriction.

Throughout this exercise, we will use the notation and results of Problem 2.11.6.

Let G be a finite group and $H \subset G$ a subgroup. We consider $k[G]$ as a $(k[H], k[G])$-bimodule (both module structures are given by multiplication inside $k[G]$). We denote this bimodule by $k[G]_1$. On the other hand, we can also consider $k[G]$ as a $(k[G], k[H])$-bimodule (again, both module structures are given by multiplication). We denote this bimodule by $k[G]_2$.

(a) Let V be a representation of G. Then, V is a left $k[G]$-module. Thus, the tensor product $k[G]_1 \otimes_{k[G]} V$ is a left $k[H]$-module. Prove that this tensor product is isomorphic to $\operatorname{Res}_H^G V$ as a left $k[H]$-module. The isomorphism

$$\operatorname{Res}_H^G V \to k[G]_1 \otimes_{k[G]} V$$

is given by $v \mapsto 1 \otimes_{k[G]} v$ for every $v \in \operatorname{Res}_H^G V$.

(b) Let W be a representation of H. Then W is a left $k[H]$-module. According to Remark 5.8.2, $\operatorname{Ind}_H^G W \cong \operatorname{Hom}_H(k[G], W)$. In other words, we have $\operatorname{Ind}_H^G W \cong \operatorname{Hom}_{k[H]}(k[G]_1, W)$. Now use part (b) of Problem 2.11.6 to conclude Theorem 5.10.1.

(c) Let V be a representation of G. Then, V is a left $k[G]$-module. Prove that not only $k[G]_1 \otimes_{k[G]} V$ but also $\operatorname{Hom}_{k[G]}(k[G]_2, V)$ is isomorphic to $\operatorname{Res}_H^G V$ as a left $k[H]$-module. The isomorphism

$$\operatorname{Hom}_{k[G]}(k[G]_2, V) \to \operatorname{Res}_H^G V$$

is given by $f \mapsto f(1)$ for every $f \in \operatorname{Hom}_{k[G]}(k[G]_2, V)$.

(d) Let W be a representation of H. Then, W is a left $k[H]$-module. Show that $\operatorname{Ind}_H^G W$ is isomorphic to $k[G]_2 \otimes_{k[H]} W$. The isomorphism $\operatorname{Hom}_{k[H]}(k[G]_1, W) \to k[G]_2 \otimes_{k[H]} W$ is given by $f \mapsto \sum_{g \in P} g^{-1} \otimes_{k[H]} f(g)$ for every $f \in \operatorname{Hom}_{k[H]}(k[G]_1, W)$, where P is a set of distinct representatives for the right H-cosets in G. (This isomorphism is independent of the choice of representatives.)

(e) Let V be a representation of G and let W be a representation of H. Use (b) to prove that $\operatorname{Hom}_G\left(\operatorname{Ind}_H^G W, V\right)$ is naturally isomorphic to $\operatorname{Hom}_H\left(W, \operatorname{Res}_H^G V\right)$.